



Sun Power Race

My name _____

Class _____

Date _____

How can sunlight make things move?

Sunlight can make electricity! A solar panel turns light into power that can spin a motor. What makes it spin faster?

My kit tick when you have it

- ☐ A small solar panel with two wires
- ☐ A small motor
- ☐ A paper flag and tape
- ☐ A sheet of baking paper for the cloud
- ☐ A book to lean the panel on

Stay safe

- Never look straight at the sun.
- Wear a hat outside.
- Ask an adult to help join the wires.

START HERE: BUILD

Build your sun spinner

1 · DO IT: TICK EACH STEP

- 1 ☐ Tape the paper flag onto the motor spindle.
- 2 ☐ Join the red wire to the red wire.
- 3 ☐ Join the black wire to the black wire.
- 4 ☐ Take the panel into the sun.
- 5 ☐ Watch the flag.

2 · LOOK & RECORD: CIRCLE WHAT YOU SEE

	How fast does the flag spin?
My sun spinner	☐ Not at all ☐ Slowly ☐ Fast

Draw your sun spinner

WORDS TO USE

solar panel motor wire spin electricity sun

TEST 1 Sun, shade or cloud?

? Where will the flag spin fastest?

1 · MY GUESS: CIRCLE ONE

In full sun

Under a "cloud"

In the shade

How sure am I?

2 · DO IT: TICK EACH STEP

- ☐ Put the panel in full sun. Watch the flag.
- ☐ Hold baking paper over the panel. Watch again.
- ☐ Move the panel into the shade. Watch again.
- ☐ Tap how fast it spun each time.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How fast does the flag spin?
Full sun	☐ Not at all Slowly Fast
Cloud	☐ Not at all Slowly Fast
Shade	☐ Not at all Slowly Fast

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right

☐ Partly right

☐ Something different happened

☐ Not sure yet

WORDS TO USE

sun

shade

cloud

light

fast

slow

power

TEST 2 Flat or facing the sun?

? Will the flag spin faster with the panel flat, or facing the sun?

1 · MY GUESS: CIRCLE ONE

☐ Flat on the ground

☐ Facing the sun

☐ Both the same

How sure am I?

2 · DO IT: TICK EACH STEP

- ☐ Lay the panel flat on the ground in the sun.
- ☐ Watch the flag. Tap how fast it spins.
- ☐ Lean it on a book so it faces the sun.
- ☐ Its shadow behind it should be as short as possible.
- ☐ Watch the flag again and tap.

3 · LOOK & RECORD: CIRCLE WHAT YOU SEE

Test	How fast does the flag spin?
Flat	☐ Not at all Slowly Fast
Facing the sun	☐ Not at all Slowly Fast

4 · WHAT HAPPENED? CIRCLE ONE

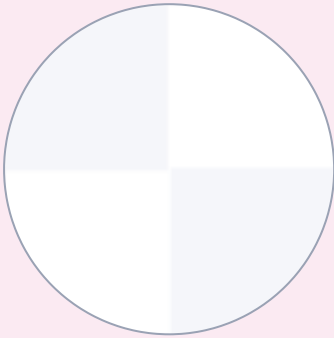
- ✔ My guess was right
- 🌐 Partly right
- 🔄 Something different happened
- 🤔 Not sure yet

WORDS TO USE

angle flat facing tilt shadow faster

🎨 Art corner: Colour spinner

Colour a circle in sections of two colours. Put it on the motor. What colour do you see when it spins?



➗ Maths corner: Fastest to slowest

Number the places 1, 2, 3 from fastest to slowest.

📖 New words

- Solar:** To do with the sun.
- Electricity:** A kind of energy that makes things work.
- Energy:** What makes things move, light up or get warm.
- Angle:** How much something is tilted.

★ What I can do now colour the stars

- I can join a solar panel to a motor.☆☆☆
- I can compare how fast it spins in sun, cloud and shade.☆☆☆
- I can say how light turns into movement.☆☆☆